VPA ICT to CS Transferability Guide

☐ VPA ICT to Computer Science Transferability Guide

Victoria Park Academy | ICT Programme → IB DP Computer Science Bridge

This document maps how ICT skills developed throughout the VPA curriculum directly prepare students for advanced Computer Science study. Each ICT skill is linked to its CS application, with grade-level progression and concrete examples.

Transferability Overview

English: The VPA ICT curriculum is intentionally designed as a foundation for Computer Science. Skills in productivity software, data management, programming, and systems thinking build progressively toward the rigour of IB DP Computer Science. This guide shows students, parents, and teachers exactly how each ICT topic transfers.

中文：VPA ICT课程被有意设计为计算机科学的基础。生产力软件、数据管理、编程和系统思维方面的技能逐步建立，为IB DP计算机科学的严谨性做好准备。本指南向学生、家长和教师展示每个ICT主题如何转移。

1. Spreadsheet Formulas → Programming Logic

Core Transfer: The logical thinking required for spreadsheet formulas directly maps to algorithmic thinking in programming. Both require understanding variables, operators, conditions, and functions.

ICT Skill	CS Application	Grade Level	Example
Simple formulas (+, -, *, /)	Arithmetic operators in Python/Java	G2-5	=A1+B1 → x = a + b
Cell references (relative/absolute)	Variable scope and memory addressing	G6-8	A1 → const int MAX = 100;
IF function	Conditional statements (if-else)	G6-8	=IF(A1>50,"Pass","Fail") → if (score > 50) { result = "Pass"; } else { result = "Fail"; }
Nested IF / IFS	Nested conditionals, switch statements	G9-10	=IF(A1>=90,"A",IF(A1>=80,"B"...)) → if-elif-else chains
VLOOKUP / HLOOKUP	Array searching, hash maps, dictionary lookup	G9-10	=VLOOKUP(X,A1:D10,2,FALSE) → dict[key] or binary search
COUNTIF / SUMIF	Loop accumulation with conditionals	G9-10	=COUNTIF(A1:A10,">50") → for i in range(10): if data[i] > 50: count += 1
Goal Seek / Solver	Algorithm optimisation, root-finding algorithms	G9-10	Goal Seek → Newton-Raphson method, binary search for solutions
Pivot Tables	Data aggregation, SQL GROUP BY, MapReduce	G9-10	Pivot Table → SELECT category, SUM(sales) FROM data GROUP BY category

ICT Skill	CS Application	Grade Level	Example
Macro recording	Procedural programming, automation scripts	G9-10	Excel Macro → Python script with loops and file I/O

Teaching Tip: When teaching spreadsheets, explicitly draw the parallel to programming. For example, after students write =IF(A1>50,"Pass","Fail"), show the Python equivalent and discuss how both evaluate a condition and branch execution.

2. Web Design → HTML/CSS → Web Development

Core Transfer: HTML and CSS skills from ICT provide the foundational markup and styling knowledge that underpin all web development, from static sites to dynamic full-stack applications.

ICT Skill	CS Application	Grade Level	Example
HTML tags (h1, p, img, a)	Semantic markup, DOM manipulation	G6-8	Title → document.createElement('h1') in JavaScript
CSS selectors and properties	Styling frameworks (Bootstrap, Tailwind), CSS-in-JS	G6-8	.class { color: red; } → styled-components in React
CSS Box Model	Layout engines, UI component design	G6-8	margin + border + padding → Flutter/Android layout constraints
Responsive design (media queries)	Mobile-first development, adaptive UI	G9-10	@media (max-width: 600px) → React Native flexbox, SwiftUI
Forms and input validation	Backend form handling, API endpoints, security	G9-10	<input type="email" required> → Express.js validation, SQL injection prevention
Multi-page site navigation	Single-page applications (SPA), routing	G9-10	<a href="page.html"> → React Router, Vue Router
Embedding media (images, video)	Content delivery networks (CDN), streaming protocols	G9-10	<img src="photo.jpg"> → CloudFront, adaptive bitrate streaming
Accessibility considerations	WCAG compliance, inclusive design, screen reader support	G9-10	alt text → ARIA labels, semantic HTML5 for assistive technology

Teaching Tip: Have students build a static website in ICT, then in IB CS extend it with JavaScript interactivity and a backend database. This creates a clear progression from markup to full-stack development.

3. Databases → SQL & Data Structures

Core Transfer: Database design skills in ICT (tables, relationships, queries) map directly to relational database theory in CS, SQL programming, and data structure concepts.

ICT Skill	CS Application	Grade Level	Example
Table design (fields, data types)	Relational schema design, data modelling	G6-8	Access table designer → CREATE TABLE statements, ER diagrams

ICT Skill	CS Application	Grade Level	Example
Primary keys	Unique identifiers, UUID generation, indexing	G6-8	AutoNumber → PRIMARY KEY, AUTO_INCREMENT, UUIDv4
Relationships (1:N, M:N)	Foreign keys, join tables, referential integrity	G9-10	Access Relationships → FOREIGN KEY constraints, junction tables
Simple queries (SELECT, WHERE)	SQL programming, database APIs	G6-8	Access Query Designer → SELECT * FROM Students WHERE Grade = 10
Sorting and filtering	ORDER BY, indexing strategies, query optimisation	G9-10	Sort A-Z → ORDER BY name ASC; CREATE INDEX idx_name
Aggregate queries (COUNT, SUM, AVG)	SQL aggregation, data analytics, BigQuery	G9-10	Query totals → SELECT AVG(score) FROM Exams GROUP BY subject
Forms for data entry	CRUD applications, RESTful API design	G9-10	Access Form → POST /api/students, form validation, CSRF tokens
Reports and grouping	Business intelligence, data visualisation, dashboards	G9-10	Access Report → Tableau, Power BI, matplotlib/seaborn
Normalisation (1NF, 2NF, 3NF)	Database design theory, redundancy elimination, ACID properties	G9-10	Splitting tables → Normal forms, Boyce-Codd Normal Form (BCNF)
Data validation rules	Input validation, schema constraints, type safety	G9-10	Field validation → CHECK constraints, JSON Schema, TypeScript types

Teaching Tip: When students design a database in Access, ask them to write the equivalent SQL CREATE TABLE statements. This bridges the gap between GUI tools and text-based programming.

4. Programming Foundations → Advanced CS

Core Transfer: Block-based programming in primary years and text-based programming in middle years build the computational thinking foundations for IB DP Computer Science algorithms, data structures, and OOP.

ICT Skill	CS Application	Grade Level	Example
Sequences (Scratch blocks)	Statement execution, control flow	G2-5	Move 10 steps, say "Hello" → print("Hello"); x = x + 10
Loops (repeat, forever)	for loops, while loops, iteration	G2-5	Repeat 10 → for i in range(10): or while count < 10:
Conditionals (if-then)	if-else, switch, ternary operators	G2-5	If touching mouse → if (distance < 10) { ... }
Variables (Scratch)	Typed variables, scope, memory allocation	G2-5	Set score to 0 → int score = 0;
Events (broadcast, when clicked)	Event-driven programming, callbacks, listeners	G2-5	When green flag clicked → button.addEventListener('click', handler)

ICT Skill	CS Application	Grade Level	Example
Python variables and I/O	Strong typing, stdin/stdout, file I/O	G6-8	name = input() → BufferedReader, Scanner, std::cin
Python lists	Arrays, ArrayLists, linked lists, dynamic arrays	G6-8	my_list = [1, 2, 3] → int[] arr = {1, 2, 3}; ArrayList<Integer>
Python functions (def)	Methods, parameter passing, return types, recursion	G6-8	def greet(name): → public void greet(String name) { }
Python dictionaries	Hash maps, hash tables, JSON objects	G9-10	student = {"name": "Amy"} → HashMap<String, String>, JSON.parse()
Python file I/O	Streams, serialization, CSV parsing, exception handling	G9-10	with open('data.txt') → FileReader, try-catch blocks, CSV libraries
Flowcharts and pseudocode	Algorithm design, UML, formal specification	G6-8	Diamond decision box → if statements, state machines
Debugging strategies	Unit testing, breakpoints, stack traces, logging	G6-8	Print variable values → debugger, JUnit, pytest, console.log

5. Systems Thinking → Computer Science Theory

Core Transfer: Understanding how hardware, software, and networks work together in ICT provides the systems-level thinking required for CS theory, architecture, and networking.

ICT Skill	CS Application	Grade Level	Example
Input → Process → Output model	Von Neumann architecture, CPU cycles	G2-5	Keyboard → CPU → Screen → fetch-decode-execute cycle
File organisation (folders)	File systems, directory trees, path resolution	G2-5	My Documents/Projects → /home/user/projects, inode structure
Storage devices (HDD, SSD, USB)	Memory hierarchy, cache, virtual memory	G6-8	SSD vs HDD → RAM vs ROM, L1/L2/L3 cache, paging
Binary and data representation	Number systems, bitwise operations, encoding	G6-8	8-bit colour → ASCII, Unicode, UTF-8, floating-point representation
Network basics (LAN, WAN, Wi-Fi)	OSI model, TCP/IP, routing algorithms	G6-8	Wi-Fi router → OSI layers 1-7, packet switching, Dijkstra's algorithm
IP addresses and DNS	Network layer, addressing schemes, NAT	G9-10	192.168.1.1 → IPv4/IPv6, subnetting, CIDR notation
Security (passwords, encryption)	Cryptography, hashing, public-key infrastructure	G9-10	Strong password → bcrypt, SHA-256, RSA, SSL/TLS handshake
Cloud storage	Distributed systems, replication, CAP theorem	G9-10	Google Drive → AWS S3, Hadoop, distributed file systems

6. Multimedia → Graphics, Vision & HCI

Core Transfer: Multimedia skills in image editing, video production, and audio processing connect to computer graphics, computer vision, and human-computer interaction in CS.

ICT Skill	CS Application	Grade Level	Example
Image resolution and pixels	Raster graphics, image processing, convolution	G6-8	1920x1080 pixels → OpenCV, image kernels, edge detection
Bitmap vs Vector	Graphics pipelines, SVG rendering, Bézier curves	G6-8	Vector graphics → PostScript, PDF rendering, game asset pipelines
Colour models (RGB, hex)	Colour spaces, image compression, dithering	G6-8	#FF5733 → HSV, CMYK, JPEG chroma subsampling
Audio sampling and bit depth	Digital signal processing, Fourier transforms, compression	G9-10	44.1kHz, 16-bit → Nyquist theorem, MP3 psychoacoustics
Video frame rates and compression	Video codecs, streaming protocols, computer vision	G9-10	30fps, H.264 → H.265/AV1, DCT, motion estimation, YOLO object detection
User interface design	Human-Computer Interaction (HCI), UX research, accessibility	G9-10	Button placement → Fitts's Law, Nielsen's heuristics, A/B testing

Progression Map: ICT → IB DP Computer Science

Year	ICT Focus	CS Readiness
G1-2	Mouse/keyboard, digital drawing, basic sequencing	Computational thinking foundations; comfort with technology
G3-5	Word processing, spreadsheets, Scratch programming, safe searching	Logic and structure; algorithmic thinking; digital literacy
G6-7	Advanced productivity, Python basics, HTML/CSS, hardware	Text-based programming; markup languages; systems understanding
G8	Databases, networks, multimedia, digital citizenship	Data structures; networking concepts; ethical reasoning
G9-10 (IGCSE)	Document production, data analysis, web authoring, advanced DB/SQL	Ready for IB DP CS: programming, data structures, systems, theory
G11-12 (IB DP)	IB Computer Science: OOP, abstract data structures, resource management, HL extension topics, IA, case study	

Key Takeaways for Students

Every ICT skill is a stepping stone to CS. The spreadsheet formula you write today is the algorithm you'll optimise tomorrow. The database you design today is the schema you'll scale tomorrow. The webpage you code today is the application you'll architect tomorrow.

给学生的关键信息：每一项ICT技能都是通往计算机科学的垫脚石。你今天写的电子表格公式就是明天要优化的算法；你今天设计的数据库就是明天要扩展的架构；你今天编写的网页就是明天要构建的应用。

Parent Guidance: Encourage students to see ICT not as "using computers" but as "learning how computers work." Ask them to explain the logic behind their spreadsheet formulas or the structure of their HTML — this metacognitive practice accelerates transfer to CS.

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